A Framework for Robust Network Traffic Prediction Using Multi-Scale Adversarial Training and an Online Adaptive Mechanism

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***Abstract*—Deep learning-based network traffic predictors are highly vulnerable to small input perturbations, which can induce significant forecast errors. To address this, we propose a defense framework that integrates multi-scale adversarial training with an online adaptive mechanism. First, multiple submodels are trained across a range of perturbation strengths to avoid overfitting to a single attack budget. Then, the online adaptive mechanism dynamically adjusts ensemble weights based on recent predictive performance. Evaluated on real-world network traffic datasets, this framework substantially reduces errors under attack while maintaining competitive accuracy on unperturbed inputs, outperforming other adversarial training methods.**

I. Introduction

Rising demand, the proliferation of sixth-generation (6G) networks and the Internet of Things (IoT), and mounting cost pressures make dynamic resource allocation in Internet access services (IAS) indispensable. Network traffic prediction (NTP) supports capacity planning and quality of service (QoS) [1]. Coarse-grained NTP is operationally central; however, forecasting errors can lead to overprovisioning (waste) or underprovisioning (Service-Level Agreement (SLA) violations) [2].

Modern NTP increasingly relies on deep learning (DL), typically assuming stable, unbiased inputs. However, adversarial manipulation of traffic (evasion/poisoning) can subtly distort volumes across routers/links and mislead predictors, while evading conventional monitoring [3]. Although adversarial attacks are well studied in vision and natural language processing (NLP) [4], their impact on coarse-grained NTP warrants further in-depth investigation.

Defenses are either *reactive* (filtering/detection) or *proactive* (training-time robustness) [5]. Reactive methods rely on detector coverage and often discard informative series, offering limited protection when perturbations resemble benign fluctuations [6]. Proactive defenses via Adversarial Training (AT) [7] face three practical limits: (i) accuracy–robustness trade-off; (ii) weak cross-budget generalization when trained at a fixed perturbation level; and (iii) no online adaptation, as defenses remain fixed during the online inference phase.

We address these gaps with a defense framework for coarse-grained NTP that couples (i) *multi-scale adversarial training*, which trains sub-models across multiple perturbation budgets to avoid overfitting to any single attack strength level, and (ii) *dynamic model averaging* (DMA), which *updates sub-model mixture weights online at inference* based on recent predictive performance. Therefore, our contributions are:

- We reveal that coarse-grained NTP is highly susceptible to subtle, single-timestep perturbations: a 12% input change can more than double prediction error ($> 100\%$), challenging the assumption of trustworthy inputs.
- We propose a defense framework that integrates multi-scale AT with a dynamic weighting, reconciling robustness across attack strengths with near-clean accuracy.
- We introduce DMA for NTP, enabling online, performance-aware reweighting of sub-models during inference under adversarial conditions.
- Across two real-world datasets, the approach consistently outperforms single-budget and stochastic-budget AT, mitigating the trade-off between accuracy and robustness while improving cross-budget generalization.

II. Literature Review

NTP forecasts future link loads to support QoS, congestion control, and capacity planning. Coarse-grained NTP aggregates traffic by station, service, or region over longer horizons [8]. Recent work, largely DL centric, has reported strong coarse-grained performance [1]. However, robustness is rarely evaluated, and most studies assume trustworthy inputs, leaving coarse-grained NTP vulnerable to adversarial manipulation.

DL models are sensitive to small, structured perturbations [3]; the Fast Gradient Sign Method (FGSM) exemplar highlights how imperceptible changes can flip predictions [7]. For NTP, such shifts can trigger over/under-provisioning and SLA violations [3]. Threats include *white-box* (full model access) and *black-box* (limited knowledge) [3], [9]. A white-box threat assumes an attacker with insider-level privileges, such as a system administrator or a hacker who has deeply compromised the network, granting them complete access to the model's architecture, parameters, and even the training data. Conversely, a black-box scenario represents a more common external threat, where the attacker has no internal privileges.

Defenses are *reactive* (detection/sanitization) or *proactive* (robustness-by-training) [5]. Reactive methods are easy to deploy but hinge on detector coverage and may discard useful data; proactive AT injects adversarial examples during training [7]. In NTP, subtle, single-digit fluctuations evade typical anomaly detectors and degrade performance [10]. Robustness thus remains underexplored relative to Independent and identically distributed (IID) skew and intrusion-detection settings.

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While evidence from related time-series forecasting domains (such as urban traffic and power grids [4], [6], [9]) shows the promise of AT, its application to the specific domain of NTP reveals three significant gaps that invite further exploration:

- **Accuracy versus robustness trade-off.** Robustness often reduces accuracy on clean data (i.e., unperturbed input), and many studies underreport strong clean baselines [11]. Since attacks are infrequent in practice, accuracy on clean data is of primary importance.
- **Perturbation generalization.** Single-budget AT generalizes poorly to unseen or mismatched perturbation magnitudes; we empirically quantify these limitations [4], [9].
- **Lack of online adaptation.** Existing proactive defenses, including ensembles, are fixed after training and do not reweight during inference. To our knowledge, this is the first work to introduce online, performance-aware reweighting at inference for coarse-grained NTP.

These gaps motivate the development of defenses that jointly preserve clean accuracy, generalize across threat levels, and adapt during inference; our work aims to address these goals.

III. Problem Formulation

We study one-step-ahead NTP: predict y_{t+1} from recent traffic to avoid SLA violations and mis-provisioning [12]. In the univariate case, given window $X_{t+1} = [y_{t-n+1}, \ldots, y_t] \in \mathbb{R}^n$, the model learns

$$f : \mathbb{R}^n \to \mathbb{R}, \qquad \hat{y}_{t+1} = f(X_{t+1}). \tag{1}$$

With auxiliary features $\mathbf{z}_t \in \mathbb{R}^d$, we form window $\mathbf{Z}_{t+1} \in \mathbb{R}^{n\times d}$ and extend to

$$f : (\mathbb{R}^n, \mathbb{R}^{n\times d}) \to \mathbb{R}, \qquad \hat{y}_{t+1} = f(X_{t+1}; \mathbf{Z}_{t+1}). \tag{2}$$

IV. Proposed Framework

A. Threat Model

We follow the threat model of [3]: an adversary controls a large botnet of compromised edge/access devices that inject subtle, coordinated perturbations into backbone or aggregation traffic (via router interfaces, aggregation switches, or gateways) rather than attempting to overwhelm links. The attack proceeds in stages: (1) an NTP model predicts future traffic for resource planning; (2) the operator allocates resources accordingly; (3) the adversary injects crafted traffic that pollutes the model's training stream; and (4) polluted data degrades forecasts, producing misallocations and cascading operational failures.

a) Attacker capabilities.: Compromise requires only a socket from IoT/edge devices (no elevated privileges), making large-scale orchestration feasible (e.g., Mirai-type botnets). Delivery vectors include preinstalled malware, malicious apps, compromised Firmware-Over-the-Air (FOTA) updates, or incentivized installs [3], [13].

b) Scope and impact.: We consider black-box, inference-time attacks whose goal is to stealthily degrade predictive accuracy (e.g., causing fraudulent bandwidth allocation or deliberate under/over-provisioning). Such attacks create substantial operational risk by inducing persistent forecasting errors and degraded QoS.

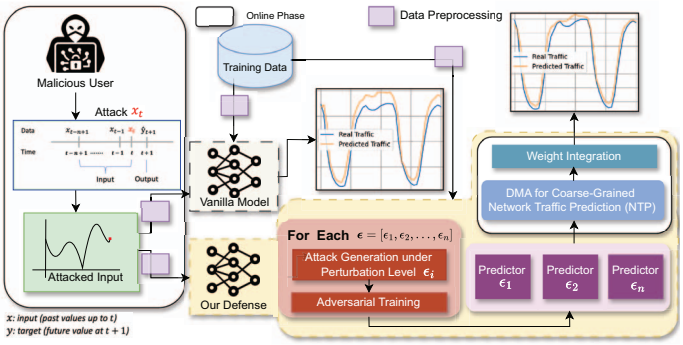


Fig. 1. Overview of adversarial attacks on traffic data and the our defense framework. Even when only a single time-point measurement (e.g., x_t) is perturbed, the Vanilla Model (no-defense baseline) suffers from large errors, leading to SLA violations and overprovisioning. In contrast, the proposed framework mitigates such vulnerabilities, ensuring robust and reliable NTP.

B. Proposed Robustness Framework

Conventional NTP models are trained on ostensibly clean histories. However, a single recent manipulation x_t can induce large next-step y_{t+1} errors [3], enabling subtle inference-time attacks that evade anomaly filters. Our framework mitigates these risks via two components:

- **Multi-ε Adversarial Training.** Train multiple submodels on FGSM-crafted examples with budgets $\{\varepsilon_1, \ldots, \varepsilon_n\}$ to avoid overfitting to a single attack magnitude and improve cross-budget generalization (cf. [4]).
- **Dynamic Model Averaging (DMA).** At inference, adaptively reweight ε-trained submodels based on recent reliability, balancing clean accuracy and robustness to black-box attacks

Operational phases (Fig. 1). *Offline:* train submodels under multiple ε using FGSM; the multi-scale design captures diverse adversarial behaviors beyond single-ε AT. *Online:* Aggregate all submodel predictions. DMA assigns weights based on stability-aware error signals to yield precise forecasts on clean inputs and resilient performance under adversarial attacks.

C. Upward-FGSM for Coarse-Grained NTP

We adapt FGSM [7] to *traffic injection*: small, sign-guided perturbations applied to the input window increase one-step prediction error in white-box attacks or harden models when used for adversarial training.

a) Core Insight.: In real networks, adversaries can typically perform only *additive* injections [3], as their access privileges are limited. Accordingly, we (i) constrain perturbations to be non-negative (*Upward FGSM*) and (ii) optionally restrict them to selected timesteps or features using a binary mask (*Masked FGSM*), both under the same threat budget.

b) Setting.: Given the length-n window

$$X_{t+1} = \big[x_{t-n+1}, \ldots, x_t\big] \in \mathbb{R}^n, \tag{3}$$

the model predicts $\hat{y}_{t+1} = F(X_{t+1})$ with loss

$$J\big(\hat{y}_{t+1}, y_{t+1}\big). \tag{4}$$

c) Standard FGSM.:

$$\eta = \epsilon \cdot \mathrm{sign}\big(\nabla_{X_{t+1}} J(\hat{y}_{t+1}, y_{t+1})\big), \qquad \bar{X}_{t+1} = X_{t+1} + \eta. \tag{5}$$

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d) Upward FGSM (additive-only).:

$$\eta_i = \begin{cases} \epsilon, & \dfrac{\partial J(\hat{y}_{t+1}, y_{t+1})}{\partial X_{t+1,i}} > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

e) Masked FGSM (localized).: For $M \in \{0,1\}^n$,

$$\eta = \epsilon \cdot M \odot \operatorname{sign}\big(\nabla_{X_{t+1}} J(\hat{y}_{t+1}, y_{t+1})\big), \qquad \bar{X}_{t+1} = X_{t+1} + \eta, \tag{7}$$

and, under the upward constraint,

$$\eta_i = \begin{cases} \epsilon, & M_i = 1 \text{ and } \dfrac{\partial J(\hat{y}_{t+1}, y_{t+1})}{\partial X_{t+1,i}} > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

D. Adversarial Training for Coarse-Grained NTP

Adversarial training (AT) [7] augments learning with FGSM-crafted "worst-case" traffic inputs, so the model minimizes loss on perturbed rather than clean data. The objective adopts the standard min–max form:

$$\min_{\theta} \ \mathbb{E}_{(X_t, y_t) \sim \mathcal{D}_{\text{train}}} \Big[\max_{\|\eta\| \le \epsilon} J\big(F_\theta(X_t + \eta),\, y_t\big) \Big], \tag{9}$$

where the inner maximization is approximated by positive-gradient FGSM. In practice, each batch is adversarially perturbed and the model updated on these examples.

E. Dynamic Model Averaging (DMA)

We employ *Dynamic Model Averaging (DMA)* as an adaptive ensemble mechanism to harden NTP against adversarial perturbations. Unlike static ensembles, DMA dynamically reweights base predictors at inference according to their recent reliability. Base learners $m_i{}_{i=1}^{k}$ are trained with diverse settings (e.g., perturbation levels ε), and DMA emphasizes those that remain accurate under the current conditions (Algorithm 1).

a) Procedure.: The ensemble is initialized with uniform weights, and each model maintains an error buffer of length W. For an incoming batch, the ensemble prediction is the weighted combination of model outputs. Once the ground truth is available, we compute the root mean squared error (RMSE) of each model, update its buffer, and form a stability-adjusted average error. Errors are then transformed into normalized scores via temperature scaling, stabilized with δ, and smoothed by momentum β to update weights online.

b) Threat model.: We consider *black-box* adversaries who cannot access or alter NTP internals and can only introduce input perturbations. White-box threats are out of scope. Under this assumption, DMA adapts weights based on observed performance, providing resilience without exposing model details.

c) Advantages.: (i) *Redundancy*: ensemble members compensate for one another, reducing variance and stabilizing predictions. (ii) *Increased attack cost*: an adversary must simultaneously contend with multiple hiddenly weighted models, complicating effective attacks in black-box settings. (iii) *Complementary robustness*: diversity among base models (e.g., trained with different ε) induces distinct failure modes, improving robustness against uncertain or unseen perturbations.

Algorithm 1 Dynamic Model Averaging (DMA). Notation: m_i is the i-th base model, $\mathbf{w} = (w_1, \ldots, w_k)$ are weights ($\sum_i w_i = 1$), $p_i = m_i(X_b)$ is prediction of m_i, errBuf$_i$ stores last W errors, $\bar{e}_i$ is its mean, $\delta > 0$ ensures stability, T is temperature, $w_{\min}$ is minimum allowed weight, and β is momentum.

Require: models $m_1, \ldots, m_k$; window W; temp T; momentum β; min $w_{\min}$; $\delta > 0$

$\mathbf{w} \leftarrow \frac{1}{k}\mathbf{1}$ ▷ Initialize uniform weights
errBuf$_i \leftarrow [\,]\ \forall i$ ▷ Initialize empty error buffers
for each batch (X_b, Y_b) **do**
  $\mathbf{p} \leftarrow \big(m_1(X_b), \ldots, m_k(X_b)\big)$ ▷ Model predictions
  $y_b^{\text{pred}} \leftarrow \mathbf{w}^\top \mathbf{p}$ ▷ Ensemble prediction
  if Y_b observed **then**
    $\mathbf{e} \leftarrow \text{RMSE}(\mathbf{p}, Y_b)$ ▷ Per-model errors
    Append e_i to errBuf$_i$; keep last W ▷ Sliding window update
    $\bar{\mathbf{e}} \leftarrow \big(\text{mean}(\text{errBuf}_1), \ldots, \text{mean}(\text{errBuf}_k)\big)$ ▷ Average error
    $\mathbf{s} \leftarrow (\bar{\mathbf{e}} + \delta)^{-1/T}$ ▷ Convert error → score, lower error = higher score
    $\tilde{\mathbf{w}} \leftarrow \mathbf{s}/\|\mathbf{s}\|_1$ ▷ Normalize scores to probability simplex
    $\mathbf{w}^{\text{new}} \leftarrow \max(\tilde{\mathbf{w}}, w_{\min})$ (elemwise) ▷ Apply minimum weight
    $\mathbf{w}^{\text{new}} \leftarrow \mathbf{w}^{\text{new}}/\|\mathbf{w}^{\text{new}}\|_1$ ▷ Renormalize to sum=1
    $\mathbf{w} \leftarrow \beta\,\mathbf{w} + (1-\beta)\,\mathbf{w}^{\text{new}}$ ▷ Momentum update
    $\mathbf{w} \leftarrow \mathbf{w}/\|\mathbf{w}\|_1$ ▷ Ensure weights sum to 1
  end if
end for
return $\{y_b^{\text{pred}}\}$ and $\{\mathbf{w}\}$ trajectory

V. Experiments

We empirically assess feasibility and robustness on real network-traffic datasets, using standard forecasting metrics (Sec. V-B). Each configuration is run 5 times and we report means. We structure four complementary cases: (I) vulnerability of standard NTP under attacks, (II) baseline defenses and their accuracy–robustness tradeoff, (III) robustness of the proposed framewrok, and (IV) comparison with stochastic-ϵ AT. Based on a preliminary study, we fix DMA hyperparameters to W=1, β=0.3, T=0.8, and $w_{\min}$=0.02, which provide a good balance between adaptation speed, stability, and robustness. Perturbations below 12% of traffic volume are assumed detector-invisible. For methods spanning multiple perturbation levels (Case III & IV), training covers $\epsilon \in \{0, 0.03, 0.06, 0.09, 0.12\}$, where ϵ=0 denotes clean data.

A. Datasets

We use two public real-world network traffic datasets: (1) **UECE**—hourly campus-backbone demand (Jan–Jul 2019, Ceará, Brazil) [14]; (2) **CERNET**—origin–destination traffic matrices from the CERNET backbone (Feb–Mar 2013) [15], aggregated into temporal sequences. In data preprocessing, Missing values are interpolated, features standardized, and data split sequentially (train/val/test: 70/15/15% for UECE,

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70/10/20% for CERNET). Each dataset is used independently to test generalization across heterogeneous regimes.

B. Evaluation Metrics

We evaluate forecasts with two metrics, where lower is better: **Weighted Mean Absolute Percentage Error (WMAPE)**, a unit-invariant relative error preferred under adversarial injection, and **Root Mean Squared Error (RMSE)**, a complementary absolute-error measure that highlights large deviations.

C. NTP Model Settings

We use a Temporal Convolutional Network (TCN) as the predictor, which has previously been shown to be effective for coarse-grained NTP [1]. The look-back window has length 8; adversarial perturbations are applied only to the most recent input (i.e., the last element of the window; see Fig. 1); and the forecast horizon is one step. To isolate adversarial effects, we intentionally de-emphasize feature engineering and model selection (though important in practice) [9]. We impose no feature constraints and adopt an NTP-standard configuration that forecasts directly from past observations [3], [12]. The same feature set is used across datasets to ensure comparability. The case studies show this setup achieves acceptable accuracy, making it a suitable testbed for robustness evaluation.

D. Input-Perturbation Settings

To simulate stealthy threats, we focus on inference-time, upward-only perturbations constrained to detector-invisible magnitudes below 0.12. This setup models an adversary who can only inject malicious traffic. Accordingly, training-time attacks and downward tampering are out of scope [3], [16]. We evaluate robustness against a baseline and two complementary adversaries designed to test different facets of model fragility:

- **Clean** ($\varepsilon = 0$): The essential baseline to measure the "cost of robustness" on unperturbed data.
- **Fixed Offset**: Represents a **worst-case magnitude** attack in upward-only context. It simulates a brute-force adversary or a persistent system fault that applies the maximum possible upward shift ($+\varepsilon\%$). This tests the model's resilience to relentless, high-magnitude distortion.
- **FGSM**: Represents an **intelligent adversary** that uses model gradients to apply the perturbation in the direction that is most harmful to the model's loss function. While its instantaneous magnitude might be less than Fixed Offset, it provides a crucial benchmark for the model's vulnerability to targeted, model-aware manipulations.

This evaluation strategy is deeply connected to our inference-time, upward-only threat model. At inference time, a model must react to immediate threats with no opportunity to retrain. The Fixed Offset attack simulates a persistent, brute-force injection, while Upward-FGSM simulates an adaptive, instantaneous exploitation of the model's current state. Within the upward-only constraint, they represent worst-case scenarios for perturbation *magnitude* and *efficiency*, respectively.

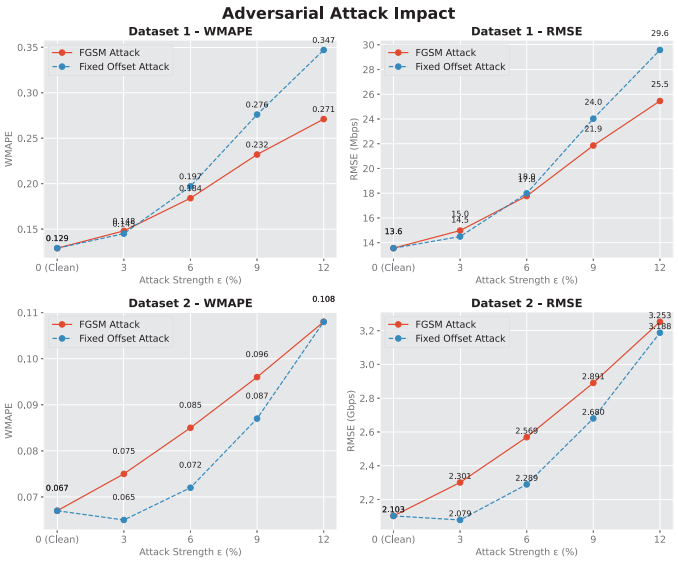


Fig. 2. WMAPE and RMSE of the undefended model (ε=0% at training) under FGSM and Fixed-Offset attacks for $\varepsilon \in \{0, 3, 6, 9, 12\}\%$ on Dataset 1 (top) and Dataset 2 (bottom).

E. Case I: No Defense Model under Adversarial Attack

This case quantifies the vulnerability of a vanilla NTP (no defense) model. As perturbation magnitude ε increases, errors rise monotonically across datasets (Fig. 2).

- *Dataset 1:* WMAPE grows from 0.129 (clean) to 0.271/0.347 (FGSM/Fixed Offset) at ε=12%; RMSE rises from 13.6 to 25.5/29.6 Mbps, i.e., +87.5%/ + 117.6%.
- *Dataset 2:* WMAPE grows from 0.067 to 0.108; RMSE from 2.103 to 3.253/3.188 Gbps at ε=12% (+54.7%/ + 51.6%).

The smaller RMSE escalation of Dataset 2 compared with Dataset 1 is consistent with its lower baseline WMAPE, indicating higher inherent accuracy. However, even a single-step perturbation (up to 12%) can more than double the RMSE—or increase it by over 50%—posing a potential threat to resource management. Both WMAPE (unit-invariant) and RMSE (large-error sensitive) reveal complementary aspects of model fragility and underscore the need for adaptive defense.

F. Case II: Single-ε Adversarial Training (AT)

We evaluate *single-ε* AT as a defense on both datasets and report clean and adversarial performance in Table I. We summarize the findings as follows:

- Clean accuracy peaks at small ε, as both datasets achieve near-best results on clean inputs with mild training noise (ε=3%), suggesting that little or no adversarial training best preserves clean accuracy.
- In Dataset 1, a robustness–accuracy trade-off is observed, where larger training budgets under strong attacks help reduce RMSE to 15–18 Mbps compared to 22–25 Mbps at $\varepsilon_{\text{train}}$=3%, but clean RMSE rises to about 16.3 Mbps (over 18% higher), reducing accuracy and stability.
- In Dataset 2, large ε attacks provide limited benefits. When $\varepsilon_{\text{train}}$=12% is used with FGSM and Fixed-Offset attacks, the model shows no significant improvement, possibly because

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TABLE I
PERFORMANCE UNDER FGSM AND FIXED OFFSET ATTACK FOR DATASET 1 (LEFT) AND DATASET 2 (RIGHT). COLUMN-WISE BEST IN **BOLD**.

Dataset 1										Dataset 2									
Train ϵ	Clean (0%)	**FGSM Attack** (Attack ϵ)				**Fixed Offset Attack** (Attack ϵ)				Train ϵ	Clean (0%)	**FGSM Attack** (Attack ϵ)				**Fixed Offset Attack** (Attack ϵ)			
		3%	6%	9%	12%	3%	6%	9%	12%			3%	6%	9%	12%	3%	6%	9%	12%
WMAPE										**WMAPE**									
3%	**0.133**	0.148	0.174	0.207	0.242	0.142	0.178	0.234	0.291	3%	**0.062**	**0.070**	0.080	0.091	0.102	**0.060**	0.066	0.079	0.098
6%	0.139	**0.145**	0.157	0.182	0.214	**0.138**	0.147	0.182	0.234	6%	0.066	0.072	0.079	0.088	0.098	0.063	0.066	0.076	0.090
9%	0.148	0.150	0.155	0.164	0.178	0.143	**0.142**	0.146	0.162	9%	0.066	0.070	**0.076**	**0.084**	**0.092**	0.062	**0.063**	**0.069**	**0.079**
12%	0.147	0.150	**0.153**	**0.159**	**0.168**	0.145	0.143	**0.144**	**0.150**	12%	0.075	0.078	0.082	0.087	0.093	0.071	0.070	0.073	**0.079**
RMSE (Mbps)										**RMSE (Gbps)**									
3%	**13.881**	**14.911**	16.803	19.490	22.515	**14.155**	16.398	20.475	25.055	3%	**1.960**	**2.164**	**2.428**	2.739	3.082	**1.956**	**2.159**	2.525	2.995
6%	14.723	15.244	**16.119**	17.877	20.297	14.304	**14.672**	16.869	20.572	6%	2.195	2.346	2.544	2.781	3.051	2.174	2.295	2.545	2.894
9%	16.290	16.536	16.903	17.459	18.388	15.537	15.005	**14.876**	15.604	9%	2.205	2.343	2.513	**2.711**	**2.931**	2.186	2.274	**2.455**	**2.712**
12%	16.397	16.675	17.007	**17.446**	**18.084**	16.060	15.684	15.386	**15.395**	12%	2.494	2.602	2.733	2.884	3.054	2.474	2.523	2.633	2.802

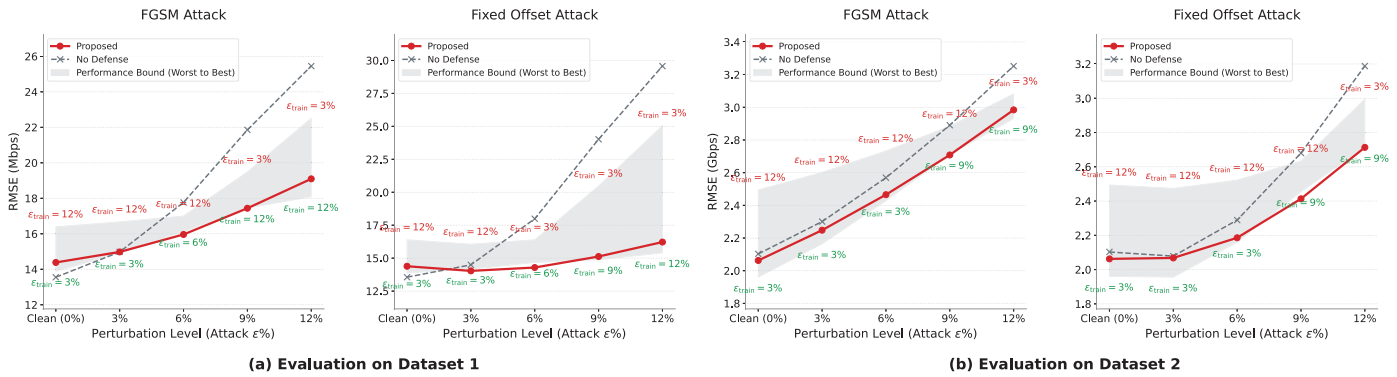


Fig. 3. Robustness comparison across two datasets. (a) Dataset 1 (RMSE in Mbps), (b) Dataset 2 (RMSE in Gbps). Each panel displays performance under FGSM (left) and Fixed-Offset (right) attacks. Red solid line: **proposed**; gray dashed line: **No Defense**. The shaded area represents the performance bound (from best to worst case across models trained with various $\varepsilon_{\text{train}}$); text labels indicate the specific training level $\varepsilon_{\text{train}}$ that achieved those bounds at each attack intensity.

the smaller dataset introduces bias and the strong baseline makes further gains less noticeable, while the increase in clean-error remains milder than in Dataset 1.

- The results indicate overfitting to the training budget, as a small $\varepsilon_{\text{train}}$ provides little protection against strong attacks, while a large $\varepsilon_{\text{train}}$ improves robustness but degrades clean accuracy. Moreover, performance under strong attacks varies, which is problematic because the strength of attacks in real-world deployment is *unknown*.

The results indicate two gaps:

1) **Clean-data degradation.** Larger $\varepsilon_{\text{train}}$ reduces vulnerability to strong attacks but substantially worsens performance on clean data, despite attacks being infrequent.
2) **Lack of generalization.** Single-ε defenses concentrate robustness near the trained budget and fail to generalize across unseen attack strengths.

Thus, a practical defense should stay robust against diverse attack strengths while preserving accuracy (see Case III).

G. Case III: Evaluation of the Proposed Framework

To overcome the generalization limits of single-ε AT while preserving clean-data accuracy, we evaluate *our proposed framework*. Figure 3 compares robustness under both attacks on the datasets against (i) *No Defense*, (ii) a *worst-case defense*, and (iii) a *best-case defense* that assumes knowledge of the perturbation level (ideal but impractical during inference).

- **Against perturbations.** For *No Defense* and the *worst case*, RMSE increases sharply as the attack strength ε grows, indicating high vulnerability. In contrast, *our proposed framework* stays close to the *best case* and sometimes demonstrates greater stability than single-ε adversarial training at medium to high perturbation levels (e.g., ε=6% on Dataset 1 and ε=9% on Dataset 2).
- **Clean-data performance.** As expected, training without adversarial samples yields the best clean results; with single-ε AT, the smallest budget (3%) is most favorable. *our proposed framework* typically trails the *best case* by a small margin and, on Dataset 2, even surpasses *No Defense* at ε=0, indicating minimal cost on clean inputs.
- **High-ε attacks.** When $\varepsilon \in 9\%, 12\%$, *our framework* shows significantly smaller increases in RMSE compared to *No Defense* and the *worst case*, staying close to the *best case* and demonstrating strong robustness under severe attacks.
- **Across datasets and attack types.** The consistent, low-variance RMSE of *our proposed framework* across both datasets and attack families demonstrates generalized robust-

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TABLE II
RMSE COMPARISON (LOWER IS BETTER). UNITS: DATASET 1 IN MBPS; DATASET 2 IN GBPS.

Dataset	Attack	ϵ (%)	No Defense	S-AT	Proposed
Dataset 1	FGSM	0 (Clean)	**13.554**	15.751	14.385
		3	**14.978**	16.266	**14.978**
		6	17.767	17.030	**15.962**
		9	21.854	18.145	**17.439**
		12	25.454	19.609	**19.097**
	Fixed	0 (Clean)	**13.554**	15.751	14.385
		3	14.494	15.480	**14.033**
		6	17.986	15.646	**14.288**
		9	24.023	16.459	**15.124**
		12	29.577	17.960	**16.221**
Dataset 2	FGSM	0 (Clean)	2.103	2.200	**2.063**
		3	2.301	2.371	**2.248**
		6	2.569	2.591	**2.465**
		9	2.891	2.849	**2.708**
		12	3.253	3.139	**2.984**
	Fixed	0 (Clean)	2.103	2.200	**2.063**
		3	2.079	2.212	**2.068**
		6	2.289	2.356	**2.186**
		9	2.680	2.612	**2.413**
		12	3.188	2.955	**2.713**

ness, maintaining performance below the error threshold even under adversarial attacks during inference.

In realistic settings with unknown and unavoidable perturbations, our framework delivers performance close to the *best case*, while maintaining competitive clean accuracy, providing practically valuable robustness.

H. Case IV: Proposed Framework vs. Stochastic-ε AT (S-AT)

We evaluate our proposed framework against Stochastic-ε Adversarial Training (S-AT), a technique designed to prevent overfitting to a specific perturbation budget. By randomly sampling the budget ε during training, S-AT aims to achieve more uniform robustness across various attack strengths.

While S-AT outperforms standard single-ε training by reducing budget-specific overfitting (as shown by comparing the results of single-ε AT in Table I with the S-AT in Table II), our framework achieves lower error across both datasets and most attack scenarios, demonstrating stronger generalization across different training budgets. Under the condition of **not requiring prior knowledge of the attack budget during inference**, our framework consistently yields lower RMSE than S-AT across all tested perturbation strengths.

I. Discussion

The proposed defense framework advances the state of adversarially robust NTP by harmonizing accuracy and robustness across unseen perturbations. The results on two real-world datasets confirm the generalization capability of the framework, demonstrating that it can be readily adapted to different operational environments. **This work thus provides a solid foundation for future research** and journal-level extensions.

Notably, the modular design allows network operators or developers to replace the base predictor with their preferred architecture while maintaining the defense pipeline, ensuring flexibility without compromising robustness. In addition, the dynamic weighting mechanism adaptively adjusts ensemble weights, preventing imbalanced resource or inference allocation even under malicious or biased traffic.

VI. CONCLUSION

We showed that coarse-grained NTP is highly vulnerable to subtle adversarial perturbations, threatening reliable resource management. To counter this, we proposed a framework combining multi-scale adversarial training with an online adaptive mechanism, achieving robust and accurate forecasts across diverse attack strengths. Experiments on two real-world datasets confirm consistent gains over single-budget and stochastic AT.

Our deployment-friendly framework advances robust time-series forecasting, enabling trustworthy, attack-resilient resource allocation in future networks.

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